

Supplementary Appendix for: Inducing Sustained Creativity and Diversity in Large Language Models*

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Appendix A Theoretical Analysis: Joint Responsiveness and Coverage Separation

Here, we formalize why RD produces *semantically structured* diversity—covering distinct conceptual regions—rather than mere noise. Throughout this appendix, P denotes a fixed user prompt.

A.1 Setting and Definitions

Let $\mathcal{R}(P)$ denote the set of relevant responses to P , partitioned into K semantic clusters $\{C_1, \dots, C_K\}$. RD draws from two finite vocabularies: a priming-phrase vocabulary $\mathcal{V}_p$ and a diverting-token vocabulary $\mathcal{V}_d$, with sampling measures μ_p and μ_d (uniform in the default RD algorithm of Section 3). Each RD run $t = 1, \dots, T$ independently draws $(p_t, d_t) \sim \mu_p \otimes \mu_d$, and the LLM produces an output Y_t with $Y_t \mid (P, p_t, d_t) \sim \text{LLM}(\cdot \mid P, p_t, d_t)$, independent across t . Under ordinary decoding (OD), no augmentation is applied: $Y_t \sim \text{LLM}(\cdot \mid P)$, again i.i.d. across runs. Throughout this appendix, Y_t is one *idea*—the unit the partition $\{C_k\}$ is defined over—and T is a budget of ideas, not of API calls. A prompt that asks for five ideas per call therefore reaches $T = 250$ in 50 calls, and it is that idea budget the calibration of Section A.5 refers to.

We make a standing *relevance* assumption: $Y_t \in \mathcal{R}(P)$ with probability $1 - \eta$ under both decoding methods, where $\eta := \mathbb{P}[Y \notin \mathcal{R}(P)] \leq 0.03$ is the (small) irrelevance probability documented empirically in Sections 4.1, 4.2, 4.3, and 4.4. Measured at the level of individual ideas—the unit the partition $\{C_k\}$ is defined over, and therefore the one relevant here—OD scores between 0.98 and 1.00 and RD between 0.94 and 1.00 across all reported settings, and Appendix Q shows that the residual RD gap is largely an artifact of the judge’s typicality bias. Scored instead at the level of complete runs (Appendix B.1), the spread is wider—RD ranges from 0.910 to 0.986 on the high-stakes prompts—because a single stem-captured generation invalidates a whole output rather than one idea; a bound stated in terms of runs would need a correspondingly larger η . Under this assumption the partition $\{C_1, \dots, C_K\}$ exhausts the support of the response distribution up to a mass of η , and $\sum_{k=1}^K q_k^{\text{OD}} = \sum_{k=1}^K q_k^{\text{RD}} = 1 - \eta$ (with q_k defined below). The structural ceiling of Theorem 1 becomes $\rho \leq (1 - \eta)/K$, and every coverage bound below carries an implicit multiplicative $(1 - \eta)$ factor of magnitude at most 0.03; for readability we treat $\eta = 0$ in the displays and absorb the at-most-0.03 correction into the residual slack between bound and observation.

The partition $\{C_1, \dots, C_K\}$ is a theoretical primitive (a fixed, conventionality-neutral decomposition of $\mathcal{R}(P)$) operationalized by the clustering pipeline of Appendix B.4; the bounds below hold for any such fixed pipeline, and Section A.5 reports numerical values for the granularity used in our main experiments.

Definition 1 (Per-pair cluster activation). *For each cluster C_k and each pair $(p, d) \in \mathcal{V}_p \times \mathcal{V}_d$, the conditional cluster probability is*

$$\pi_k(p, d) := \mathbb{P}_{Y \sim \text{LLM}(\cdot \mid P, p, d)}[Y \in C_k].$$

The induced per-run cluster activation rate is

$$q_k^{\text{RD}} := \mathbb{E}_{(p, d) \sim \mu_p \otimes \mu_d}[\pi_k(p, d)], \quad q_k^{\text{OD}} := \mathbb{P}[Y \in C_k \mid P].$$

Definition 2 (Semantic Coverage). *The coverage of a decoding method after T independent runs is*

$$\text{Cov}(T) = \frac{1}{K} |\{k : \exists t \leq T, Y_t \in C_k\}|.$$

A.2 Two Empirically Grounded Assumptions

The coverage separation rests on two assumptions, one about each decoding method. The first says OD’s probability mass piles onto a small set of favored clusters, leaving the long tail of relevant clusters with negligible mass. The second says RD’s two perturbations — the priming phrase and the diverting token — each, on their own, expose every cluster.

Assumption 1 (Modal Concentration). *Under OD, almost all probability mass concentrates on a small set S of $K_0 \ll K$ modal clusters, with the remaining clusters sharing at most ε total probability:*

$$|S| = K_0, \quad \sum_{k \notin S} q_k^{\text{OD}} \leq \varepsilon.$$

In words: of the K relevant clusters, OD concentrates on K_0 favorites (the *modal set* S); the remaining $K - K_0$ clusters together pick up at most ε probability per draw. Sampling from such a distribution returns the modal clusters repeatedly while almost never reaching the tail. Newer, higher-accuracy LLMs sharpen this concentration further (Section 4.5.1, Divekar et al. (2024)); Section A.5 measures $K_0 = 34$ and $\varepsilon = 0.248$ on our brainstorm benchmark.

The second assumption captures how RD’s two perturbations reach clusters OD misses. RD samples both a priming phrase p and a diverting token d at every run (Algorithm 1). The assumption is that, for every cluster C_k , a non-negligible fraction of priming phrases push the LLM toward C_k on their own, *and* likewise for diverting tokens — either perturbation alone suffices to activate C_k , with no requirement that both align.

Assumption 2 (Joint Responsiveness). *There exist constants $\gamma_p, \gamma_d \in (0, 1]$ and $\beta \in (0, 1]$ and, for every cluster C_k , subsets $\mathcal{V}_p^{(k)} \subseteq \mathcal{V}_p$ and $\mathcal{V}_d^{(k)} \subseteq \mathcal{V}_d$ with $\mu_p(\mathcal{V}_p^{(k)}) \geq \gamma_p$ and $\mu_d(\mathcal{V}_d^{(k)}) \geq \gamma_d$, such that*

$$\pi_k(p, d) \geq \beta \quad \text{whenever } p \in \mathcal{V}_p^{(k)} \text{ or } d \in \mathcal{V}_d^{(k)}.$$

In words: $\mathcal{V}_p^{(k)}$ is the priming-phrase strip favoring C_k (sampling mass at least γ_p), $\mathcal{V}_d^{(k)}$ is the analogous diverting-token strip (mass at least γ_d), and β is the activation strength once either favorable perturbation is drawn. The key structural point is that the activating set is the *union* of the two strips, not their intersection: a single favorable knob already triggers C_k . Because p and d are sampled independently under $\mu_p \otimes \mu_d$, inclusion–exclusion gives an activating-set measure of $\gamma_p + \gamma_d - \gamma_p \gamma_d$, so

$$q_k^{\text{RD}} = \mathbb{E}_{(p,d)}[\pi_k(p, d)] \geq (\gamma_p + \gamma_d - \gamma_p \gamma_d) \beta.$$

This lower bound is *uniform across clusters*: every C_k , modal or tail, has per-run RD activation probability at least $(\gamma_p + \gamma_d - \gamma_p \gamma_d) \beta$. Modal Concentration imposes no analogous floor on OD — tail clusters under OD can have q_k^{OD} arbitrarily small. The contrast between OD’s missing floor and RD’s uniform floor is what drives the coverage separation in the next subsection.

A.3 Theorem: Coverage Separation

Theorem 1 (Coverage Separation). *Suppose Assumptions 1 and 2 hold, and let $\rho := (\gamma_p + \gamma_d - \gamma_p \gamma_d) \beta$. Then for every $T \geq 1$,*

$$\mathbb{E}[\text{Cov}_{\text{OD}}(T)] \leq \frac{K_0}{K} + \frac{T\varepsilon}{K}, \quad (1)$$

$$\mathbb{E}[\text{Cov}_{\text{RD}}(T)] \geq 1 - e^{-\rho T}. \quad (2)$$

For any target coverage level $1 - \delta$ with $\delta \in (0, 1 - K_0/K)$, the upper bound (1) on $\mathbb{E}[\text{Cov}_{\text{OD}}(T)]$ first reaches $1 - \delta$ at

$$T \geq T_{\text{OD}}^{\text{cert}}(\delta) := \frac{(1 - \delta)K - K_0}{\varepsilon},$$

while the lower bound (2) on $\mathbb{E}[\text{Cov}_{\text{RD}}(T)]$ certifies $\geq 1 - \delta$ as soon as

$$T \geq T_{\text{RD}}^{\text{cert}}(\delta) := \ln(1/\delta)/\rho.$$

(These thresholds are bound-driven: $T_{\text{OD}}^{\text{cert}}$ is when the OD upper bound stops ruling out $1 - \delta$; $T_{\text{RD}}^{\text{cert}}$ is when the RD lower bound rules it in.) Disjointness of $\{C_k\}$ together with the relevance assumption forces $\sum_k q_k^{\text{RD}} = 1$, so the uniform lower bound $q_k^{\text{RD}} \geq \rho$ implies $\rho \leq 1/K$. In the saturating regime $\rho = \Theta(1/K)$, the speedup factor

$$T_{\text{OD}}^{\text{cert}}/T_{\text{RD}}^{\text{cert}} = \Omega\left(\frac{K\rho}{\varepsilon \ln(1/\delta)}\right) = \Omega\left(\frac{1}{\varepsilon \ln(1/\delta)}\right)$$

is governed by ε , not by K . Equivalently, at any fixed run budget T , the coverage gap

$$\mathbb{E}[\text{Cov}_{\text{RD}}(T)] - \mathbb{E}[\text{Cov}_{\text{OD}}(T)] \geq (1 - e^{-\rho T}) - (K_0 + T\varepsilon)/K$$

is order-1 at $T \asymp 1/\rho$: OD's upper bound stays at $K_0/K + \varepsilon/(K\rho) \asymp K_0/K + \varepsilon$ while RD's lower bound reaches $1 - 1/e \approx 0.63$.

Lemma 1 (OD upper bound). *Under Assumption 1, for every $T \geq 1$, $\mathbb{E}[\text{Cov}_{\text{OD}}(T)] \leq K_0/K + T\varepsilon/K$.*

Proof. For each cluster C_k , let $H_k := \mathbb{1}\{\exists t \leq T : Y_t \in C_k\}$. For $k \in S$, bound trivially $\mathbb{P}[H_k = 1] \leq 1$. For $k \notin S$, the union bound across the T i.i.d. runs gives $\mathbb{P}[H_k = 1] \leq \sum_{t=1}^T \mathbb{P}[Y_t \in C_k] = Tq_k^{\text{OD}}$. Hence

$$\mathbb{E}[\text{Cov}_{\text{OD}}(T)] = \frac{1}{K} \sum_{k=1}^K \mathbb{P}[H_k = 1] \leq \frac{1}{K} \left(\sum_{k \in S} 1 + \sum_{k \notin S} Tq_k^{\text{OD}} \right) \leq \frac{K_0}{K} + \frac{T\varepsilon}{K},$$

using $|S| = K_0$ for the first sum and Assumption 1 for the tail. ■

Lemma 2 (RD lower bound). *Under Assumption 2, with ρ as in Theorem 1, for every $T \geq 1$, $\mathbb{E}[\text{Cov}_{\text{RD}}(T)] \geq 1 - e^{-\rho T}$.*

Proof. The triples (p_t, d_t, Y_t) are i.i.d. across t by construction, so the events $\{Y_t \in C_k\}_{t=1}^T$ are i.i.d. Bernoulli with marginal probability $\mathbb{P}[Y_t \in C_k] = \mathbb{E}_{(p,d)}[\pi_k(p, d)] = q_k^{\text{RD}}$. Therefore

$$\mathbb{P}[H_k = 0] = (1 - q_k^{\text{RD}})^T \leq (1 - \rho)^T \leq e^{-\rho T},$$

where the first inequality uses Assumption 2 to bound $q_k^{\text{RD}} \geq \rho$ uniformly in k , and the second uses $1 - x \leq e^{-x}$ for $x \in [0, 1]$. Hence $\mathbb{P}[H_k = 1] = 1 - \mathbb{P}[H_k = 0] \geq 1 - e^{-\rho T}$ for every k , and averaging gives

$$\mathbb{E}[\text{COV}_{\text{RD}}(T)] = \frac{1}{K} \sum_{k=1}^K \mathbb{P}[H_k = 1] \geq 1 - e^{-\rho T}. \quad \blacksquare$$

Proof of Theorem 1. Inequality (1) is Lemma 1 and inequality (2) is Lemma 2. The certification thresholds follow by inverting each display: requiring the OD upper bound $K_0/K + T\varepsilon/K \geq 1 - \delta$ rearranges to $T \geq ((1 - \delta)K - K_0)/\varepsilon$ (well-defined when $\delta < 1 - K_0/K$), and requiring the RD lower bound $1 - e^{-\rho T} \geq 1 - \delta$ rearranges to $T \geq \ln(1/\delta)/\rho$. The constraint $\rho \leq 1/K$ uses the relevance assumption ($\sum_k q_k^{\text{RD}} = 1$) together with $q_k^{\text{RD}} \geq \rho$. $\blacksquare$

A.4 Connection to the Marginal Ablations

The framework subsumes the priming-only and diverting-only ablations of Section J as boundary cases: fixing d_0 collapses Assumption 2 to a one-channel statement with rate $\gamma_p\beta$ (and symmetrically $\gamma_d\beta$ for the diverting-only setting), each strictly below the joint rate $\rho = (\gamma_p + \gamma_d - \gamma_p\gamma_d)\beta$ whenever $\gamma_p, \gamma_d \in (0, 1)$. This predicts joint RD dominates either marginal, as observed in Figure A5.

A.5 Empirical Calibration of ρ, ε , and K_0

The bound in Theorem 1 is most informative once its parameters are pinned to actual numbers. We calibrate (K_0, ε, ρ) *directly from data*, by reading the empirical cluster-mass distributions q_k^{OD} and q_k^{RD} off the brainstorm replication archive and the diverting-token responsiveness data of Section K. The partition $\{C_k\}$ is the embedding-cosine clustering at threshold 0.73 of Appendix B.4 (the brainstorm setting used in the main figures and in Table A7). All medians below are across the 50 GPT-5.1 brainstorm topics, each generated at a budget of $T = 250$ ideas (50 calls $\times$ 5 ideas); the per-topic CSVs and the clustering script are released with the replication archive (coverage_calibration/).

Directly measured (K_0, ε) from the OD cluster-mass distribution. For each topic, sort $\{q_k^{\text{OD}}\}$ in decreasing order and define $K_0(\tau)$ as the smallest set whose cumulative mass exceeds τ and $\varepsilon(\tau) := 1 - \sum_{k \in S} q_k^{\text{OD}}$ as the residual tail mass. At the natural choice $\tau = 0.75$ (three-quarters of OD’s mass concentrated on the modal set), we measure

$$K_0 = 34 \quad (\text{IQR } [18, 70]), \quad \varepsilon = 0.248 \quad (\text{IQR } [0.244, 0.248]).$$

Across topics, ε tightens for newer models (Gemini-3.1-Pro, GPT-5.1) relative to GPT-3.5 and Deepseek-3, in line with the trend documented in Section 4.5.1. The aggregate

quantity that enters the OD upper bound at $T = 250$ is $K_0 + 250\varepsilon$, whose median across topics is 95.1; each topic’s own K_0 and ε are combined before the median is taken, which is why this is not $34 + 250 \times 0.248$. Hence $K \cdot \mathbb{E}[\text{Cov}_{\text{OD}}(T)] \leq 95.1$ — consistent with the OD3 cluster count of ~ 90 in Table A7.

Directly measured ρ from the RD cluster-mass distribution. The uniform lower bound $q_k^{\text{RD}} \geq \rho$ that drives the RD bound is directly observable as $\rho_{\min} := \min_k q_k^{\text{RD}}$. Across the 50 topics,

$$\rho_{\min} = 0.0040 \quad (\text{IQR } [0.0040, 0.0040]),$$

which equals $1/T = 1/250$: the worst-case cluster is one reached by exactly one idea across the entire budget. The structural ceiling $1/K$ from Theorem 1 sits at the same value (median $1/K = 0.0040$, IQR $[0.0040, 0.0041]$, since K has median 249 and IQR $[244, 250]$). The two parameters *coincide* on this calibration — $\rho_{\min} \approx 1/K$ to four significant figures — placing us in the saturating regime $\rho = \Theta(1/K)$.

Predicted vs. observed coverage at $T = 250$. Substituting the directly measured parameters into the bounds of Theorem 1 and reading observed coverage off the cumulative-distinct-cluster trajectory of each topic, we find (medians across 50 topics; for each topic we use the empirical-support proxy $K = \max(K_{\text{OD}}, K_{\text{RD}})$, where K_{OD} and K_{RD} are the two methods’ own cluster counts over their full generated corpora, which run past the $T = 250$ budget wherever a method overshoots the five-idea instruction; K equals K_{RD} on 49 of the 50 topics, the exception being one topic on which OD3 happens to produce more distinct cosine-clusters than RD3):

	predicted	observed
$\text{Cov}_{\text{OD}}(250) \leq (K_0 + 250\varepsilon)/K$	0.366	0.343
$\text{Cov}_{\text{RD}}(250) \geq 1 - e^{-T\rho_{\min}}$	0.632	1.000

The OD upper bound holds in 49/50 topics (the one failure is a 1.4% slack reversal at $\tau = 0.75$, within the choice-of- τ noise — at $\tau = 0.90$ the bound holds for all 50). The RD lower bound at $\rho = \rho_{\min} = 1/K$ holds in 50/50 topics, with median slack 0.37. That last figure should be read with care: because K is taken from the methods’ own empirical supports, $\text{Cov}_{\text{RD}}(250)$ is pinned at exactly 1.000 on the 33 topics where RD’s whole partition fits inside the 250-idea budget, and only the remaining 17 topics—those on which RD overshoots and its support exceeds the budget—put the RD side of the bound to a genuine test (observed Cov_{RD} falls as low as 0.700 there, still above the 0.632 bound). The informative half of the comparison is therefore the OD side and the gap. The predicted lower bound on that coverage *gap* is $0.632 - 0.366 = 0.27$ at $T = 250$, exceeded by the observed gap of $1.000 - 0.343 = 0.66$ by a factor of $\sim 2.4\times$. Figure A1 plots the per- T trajectories alongside the calibrated bound curves: the OD upper bound sits at or above observed Cov_{OD} across all T , and the RD lower bound sits below observed Cov_{RD} across all T , with both bounds tightest in the saturating regime $T \asymp K \approx 250$.

What the calibration shows. Substituting the measured $(K_0, \varepsilon, \rho) = (34, 0.248, 1/K)$ into Theorem 1 yields a ~ 0.27 coverage-gap lower bound at $T = 250$, exceeded by the

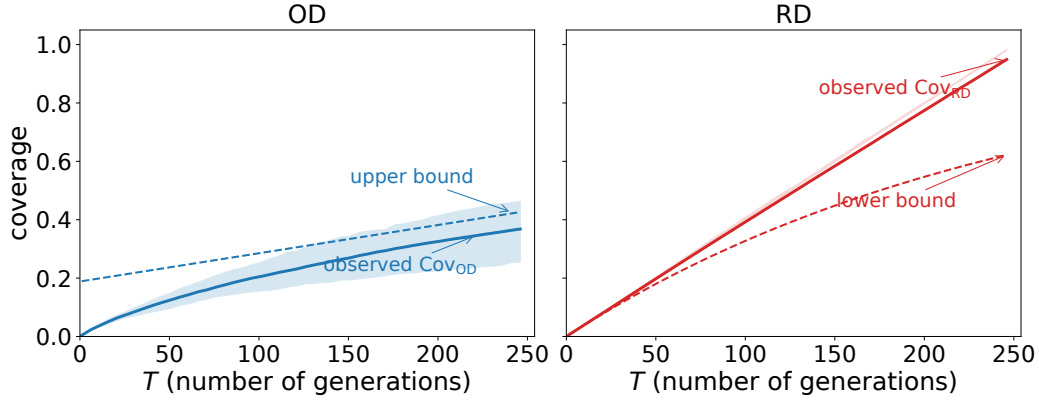


Figure A1: Coverage-separation theorem: observed vs. calibrated bounds on the 50 GPT-5.1 brainstorm topics under the cosine-similarity partition at $\tau = 0.73$. Left: ordinary decoding; the predicted upper bound $(K_0 + T\varepsilon)/K$ at $\tau = 0.75$ (dashed) sits at or above the observed Cov_{OD} trajectory (solid, mean across topics; shaded IQR) for all $T \in \{1, \dots, 250\}$. Right: recoding-decoding; the predicted lower bound $1 - e^{-T\rho_{\min}}$ at $\rho_{\min} = \min_k q_k^{\text{RD}} = 1/K \approx 0.004$ (dashed/dotted, coincident on this calibration) sits below the observed Cov_{RD} trajectory at every T .

observed RD/OD contrast on all 50 topics, and reproduces the dashed-vs-solid shape of Figure 7(a.1) without further tuning. The remaining slack between the RD lower bound and observed RD coverage reflects the bound’s worst-case character: it uses only $q_k^{\text{RD}} \geq \rho$, while in practice most clusters have q_k^{RD} several times $\rho_{\min}$.

Appendix B Evaluation Metrics for Relevance, Feasibility, Literature Grounding, Diversity, and Creativity

This appendix defines the five metrics used in the paper: relevance (Appendix B.1); feasibility (Appendix B.2); literature grounding (Appendix B.3); diversity (Appendix B.4); and creativity (Appendix B.5).

B.1 Relevance

We assess *relevance* using a LLM-judge (GPT-4o, temperature 0) Zheng et al. (2023). For each method we sample 10 runs per prompt: $n = 500$ runs per method on the 50 high-stakes topics (6,000 runs in total across the 12 methods), $n = 500$ per method on the 50 brainstorming topics, and $n = 1,000$ per method-dataset on the five 100-prompt public datasets. The same protocol and the same judge are used throughout, so relevance is comparable across the three suites.

The judge prompt is written to neutralize typicality bias. LLM judges systematically favor typical, mainstream responses over equally valid but unfamiliar alternatives, a bias documented in the training data of both judges and reward models Kirk et al. (2024); Sorensen et al. (2024); Ye et al. (2025); audits of LLM-as-a-judge systems formalize it as

“bandwagon” or popularity bias, in which judges award credit to answers that read as the consensus view and downgrade equally on-topic answers that do not [Ye et al. \(2025\)](#).

The prompt is attached below; `{prompt}` is the task prompt with its response-format instructions removed, and `{idea}` is the complete run.

Judge whether the following idea addresses the prompt. An idea may be unconventional, contrarian, or a minority position and still be relevant. Do not judge whether the idea is well-argued or likely to succeed—only whether it is on topic.

Do not penalise an idea for being unfamiliar, niche, or outside the mainstream framing of the topic. For example, for the prompt “Tell me a world history book topic,” both “medieval Church funerary practices” and “Tibetan sky burials” are fully relevant: both are important religious funerary practices, and the second is not less relevant merely because it is less prominent in mainstream discourse. Likewise, an idea that addresses a secondary aspect, or an unusual angle of the prompt is on topic. Answer Yes/No.

Prompt:

`{prompt}`

Idea:

`{idea}`

We validated the judge against human annotators on the 500-response sample described in Appendix [Q](#), which is measured at the same unit: each item shows a complete five-idea run, exactly what the judge scores. Running the judge prompt above on the same 500 responses, the judge and the human majority agree on 489 of 500 items (97.8%), with inter-annotator agreement of Fleiss’ $\kappa = 0.50$.

B.2 Feasibility

Besides the relevance judge, we adopt an additional feasibility judge to measure the quality of the generated idea to make sure the generated idea can be carried out in practice given the user prompt. We evaluate feasibility on exactly the same runs used to evaluate relevance—10 runs for each of 50 prompts, for a total of $n = 500$ runs per method—using the same judge (GPT-4o at temperature 0). We report it for the high-stakes suite (Table [A2](#)) and the brainstorming suite (Table [A6](#)); we do not report it for the five public datasets, whose prompts ask for essays, stories and open-ended discussion rather than for courses of action.

Judge whether the following idea is feasible for the prompt.

Judge only whether it CAN be done. Do not judge whether it is wise, likely to succeed, or likely to be chosen. An idea may be unconventional, contrarian, controversial, politically costly, or a minority position and still be entirely feasible; being unusual or unpopular is not the same as being impossible. Do not penalise an idea for being unfamiliar, niche, or outside the mainstream framing of the topic, or for lacking implementation detail. For example, for

the prompt “How should a government respond to a coastal flooding crisis,” both “expand the existing levee network” and “pay residents to permanently relocate inland” are feasible: the second is politically harder and less commonly proposed, but governments have the legal authority and the budgetary instruments to do it, so it is not less feasible merely because it is less prominent in mainstream discourse.

Answer Yes/No.

Prompt:

{prompt}

Idea:

{idea}

B.3 Literature Grounding

To assess whether a relevant idea is substantive, we propose a literature-grounding check: for each generated idea, we ask whether the mechanism it proposes is actually examined in the peer-reviewed scholarly record. This check applies whenever outputs make substantive claims that can be tied to an existing literature, and it is our strictest metric, since an idea can be on topic and feasible yet still unsupported by academics.

We evaluate grounding on the ideas that survive the relevance and feasibility filter. To avoid paying for redundant evaluations, we first collapse near-identical ideas via embedding-based cosine clustering at a deliberately tight threshold ($\tau = 0.88$, which leaves most clusters as singletons) and carry forward one representative per cluster. Each cluster representative is then evaluated by a literature-scout agent, an Anthropic Claude Opus model with live web search. The agent first identifies the general mechanism the idea instantiates, then searches the scholarly record for that mechanism, applying a single binary criterion: an item is *grounded* if at least one peer-reviewed journal article or university-press book examines the mechanism—in the case at hand or in any other setting, region, period, or dyad—and *ungrounded* otherwise. The evidentiary bar is strict—only peer-reviewed journals and university-press books count, while war-college journals, think-tank and policy reports, wargames, preprints, and journalism do not. The agent verifies that each cited work exists via live web search before recording it, and is instructed to return *ungrounded* rather than manufacture evidence. Each representative’s verdict is then propagated back to every member of its cluster, so all reported grounding rates are computed over individual ideas rather than clusters. We audit the resulting citations in two ways: every citation is resolved against Crossref to confirm the work exists, and every venue is checked against the evidentiary bar above. No fabricated reference and no disqualified venue appears in the final set. The complete prompts, verdicts, and verified citations are released with the replication archive.

B.4 Diversity

We quantify *diversity* by extracting ideas using GPT-4o with structured output (which is highly accurate because outputs are arranged in bullet points) and computing the cumulative number of unique clusters generated at each run. That is, by running the clustering

algorithm on all responses produced up to a given point (Run i), we record the unique cluster count and plot this growth against the run index. We apply 10 distinct clustering algorithms. As we observe high consistency across them all (see Tables A3, A5 and A7 for the per-algorithm numerical comparison), we present in the paper only the most commonly used metric, embedding-based cosine similarity.

We also employ an LLM-judge to evaluate pairwise diversity. The diversity-judge prompt is pasted below. Diversity is assessed using 20 randomly sampled response pairs per model-prompt-method combination on a three-point scale: almost identical (0 point), partially similar (1 point), and mostly different (2 points), in which scale’s order is shuffled to avoid position bias. The prompt is below. “user prompt” is the prompt used to generate the responses. “response0” and “response1” are the chosen pairs to be compared. The normalized mean score represents the method’s diversity level.

You are an AI assistant. Your task is to evaluate the similarity between two passages based on the user prompt provided. Carefully consider and compare the following aspects: 1) Concepts presented, 2) Writing style, 3) Tone of voice, 4) Perspectives, and 5) Opinions.

- For creative writing, pay close attention to the story line. If they are different, then classify as “Mostly Different”.
- For argumentative essay, pay close attention to the arguments, logic and examples used. If these elements are different, then classify as “Mostly Different”.
- For history and science questions, pay close attention to the concepts, perspectives, opinions, and the tone used.

If these elements are different, then classify as “Mostly Different”. After analyzing, provide a brief explanation of your reasoning. Then, classify the passages into one of these categories: “{scale[0]}”, “{scale[1]}”, “{scale[2]}”,

User prompt: {user prompt}

Passage 1: {response0}

Passage 2: {response1}

Our clustering methods include (1) Embedding-based Cosine Similarity; (2) TF-IDF (Term-Frequency-Inverse-Document-Frequency)-based Cosine Similarity; (3) Embedding-based Hierarchical Agglomerative Clustering (HAC) A; (4) Embedding-based Hierarchical Clustering B; (5) Graph-based Spectral Clustering; (6) Leiden Community Detection; (7) BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies); (8) DeBERTa model for similarity classification from NoveltyBench [Zhang et al. \(2025\)](#); (9) BERTscore partition from NoveltyBench [Zhang et al. \(2025\)](#); (10) Unigram partition from NoveltyBench [Zhang et al. \(2025\)](#). We apply the same clustering threshold τ across all methods to ensure a fair comparison. The value of τ is chosen carefully so that it meaningfully differentiates performance between methods. Code is in our replication archive.

B.5 Creativity

We define *creativity* as the degree to which one distribution of ideas occupies novel regions in the embedding space not covered by the other. To measure this, we first cluster each set of response embeddings using cosine-similarity clustering with a threshold of 0.73 (0.83 for the five public datasets as the answers are longer): embeddings that are sufficiently similar are grouped together, and each cluster is represented by its centroid. We then assess *coverage* between two sets of centroids using a nearest-neighbor criterion. A centroid from one distribution is considered “covered” by the other if its nearest neighbor in the other set falls within an adaptive distance threshold, set as the 95th percentile of the internal nearest-neighbor distances within the reference set. This adaptive threshold naturally accounts for differences in cluster density across prompts and models, avoiding the need to hand-tune a fixed distance cutoff. We report the percentage of covered centroids, averaged over 50 bootstrap iterations at each evaluation point. The shaded region in the plot represents the interquartile range (25th–75th percentile) across prompts, indicating where the middle 50% of individual coverage curves fall.

Appendix C Models and Baseline Methods

We evaluate RD against several prompt-engineering alternatives on two suites: 50 high-stakes decision-making prompts from Appendix G and 50 brainstorming topics from Appendix H.1. RD and OD are evaluated with four LLMs, listed in increasing order of performance on fact-based benchmarks: Deepseek-3, GPT-3.5, GPT-5.1, and Gemini-3.1-Pro. Thus, OD1–OD4 and RD1–RD4 denote ordinary decoding and recoding decoding, on these four models. All prompt-engineering baselines use GPT-3.5 and carry the suffix 2 when the model identifier is included. We hold the baseline comparison at GPT-3.5 because GPT-3.5 is available through a real Completion API, whereas the newer models can only be tested against the simulated version (see Appendix M for comparison between real Completion API and simulated Completion API.)

We evaluate the following methods:

- RD: We run the original prompt 50 or 1,000 times using the configuration described in Section 3.
- OD: We run the original prompt independently 50 or 1,000 times.
- OD with temperature of 1.6 (OD₁₆): We use the original prompt under temperature = 1.6, and then post-processing the raw output with a grammar corrector from Appendix O.
- OD with a single engineered phrase (OD_s): We append the phrase “Think outside the box. ” to the original prompt and run the resulting prompt 50 times.
- OD with multiple engineered phrases (OD_m): We construct 50 prompt-engineering phrases, including “Think outside the box,” “Please be diverse,” and “Give me unknown ideas.” On each run, we prepend one phrase to the original prompt. The complete list of phrases is provided in our replication archive.

- OD with chat history (OD_h): We use the original prompt for the first run. For each of the subsequent 49 runs, we include the complete preceding chat history together with the instruction “Generate 5 more ideas.”

Since RD outputs are post-processed using a grammar corrector, we include an ablation study in Appendix L that evaluates five OD variants with different high-temperature settings. All variants are post-processed using the same grammar corrector. Among them, OD_{16} achieves the best aggregate performance in terms of relevance and diversity.

Appendix D Battlefield Implementation Details

Here, we provide details for the analysis in Section 4.1. First, we set a maximum output token length of 150 for each run and prepend the substantive prompt “Respond in bullet points. Do NOT include sub-bullet points.” We then use GPT-4o with structured output to extract the battlefields mentioned. Next, we use Gemini-2.5 Flash with search retrieval to fact-check the extracted battlefields, obtain their corresponding geolocations, and remove duplicates. Overall, RD and OD produce 113 and 0 irrelevant battlefields out of 5,000 extracted ones, resulting in relevance scores of 0.98 and 1.00, respectively, under the relevance metric of Appendix B.1. And Gemini identified 22 and 0 invented place names from RD and OD, respectively. We remove irrelevant and invented battlefields before visualization.

Appendix E Game-Theoretic Analysis Implementation Details

The three scenarios are elicited with the prompts below. Every prompt follows the same three-part template: the scenario, the baseline extensive-form tree as specified in Morrow (1994), and the instruction to list changes to that baseline.

Deterrence (US–China over Taiwan).

“You are a game theorist constructing an extensive-form deterrence game to analyze the China–United States brinkmanship over Taiwan. The baseline tree is a pure ladder of decisions in which a challenger (China) chooses to challenge or hold; if challenged, the defender (United States) resists or concedes; and if resisted, the challenger presses the threat or backs down, with terminals ordered status quo, concessions, challenger-backs-down, and war. List 5 different changes to the baseline tree—this can include new players, feasible actions, moves by nature (e.g., uncertainty about resolve or capabilities), or payoff parameters—that would meaningfully affect the analysis of deterrence in this interaction. Respond in bullet points. Do NOT include sub-bullets. Limit each point to 10 words.”

Bargaining (US–Russia strategic arms control).

“You are a game theorist constructing an extensive-form alternating-offers bargaining game to analyze US–Russia strategic arms control. The baseline tree is an alternating sequence of offers in which the United States proposes a division of warhead

and launcher ceilings; Russia either accepts the offer or rejects it and submits a counteroffer in the next period, with the proposer role alternating each round and a common discount factor eroding both players’ payoffs over time. List 5 different changes to the baseline tree—this can include new players, feasible actions, moves by nature (e.g., uncertainty about reservation values or patience), or payoff parameters—that would meaningfully affect the analysis of bargaining in this interaction. Respond in bullet points. Do NOT include sub-bullets. Limit each point to 10 words.”

Signaling (Iran nuclear standoff).

“You are a game theorist constructing an extensive-form crisis-signaling game of incomplete information to analyze the Iran nuclear standoff. The baseline tree begins with a root-level move by Nature that determines whether the sender (Iran) is type 1, a determined proliferator intent on weaponization, or type 2, a restrained status-quo actor content with a latent civilian capability; Iran then chooses whether to build or restrain, and the receiver (United States)—observing Iran’s costly signal but not Nature’s initial draw—responds by acquiescing or striking at a partitioned information set that pools the structurally distinct nodes it cannot independently verify. List 5 different changes to the baseline tree—this can include new players, feasible actions, moves by nature (e.g., uncertainty about resolve or breakout capability), or payoff parameters—that would meaningfully affect the analysis of signaling in this interaction. Respond in bullet points. Do NOT include sub-bullets. Limit each point to 10 words.”

The outputs are then processed in four steps. First, we use an LLM relevance judge to evaluate OD and RD outputs and discard those judged irrelevant, so that every downstream step operates only on the relevant ones (Appendix B.1). Averaged over the three prompts, the relevance judge scores OD at 1.00 and RD at 0.99. We also apply the feasibility judge on the same runs. Both methods score 1.000. Second, we decompose each relevant output into the structural elements of an extensive-form game with an LLM extractor. We use an LLM extractor to extract the element and assigns it to one of the primitive categories (*player*, *action*, *nature*, or *payoff*). Because the prompts instruct the model to answer in bullet points, the extraction is highly accurate. Third, we deduplicate the extracted elements within each primitive category by embedding-based cosine clustering, reporting the unique-cluster count as the diversity measure (Appendix B.4). Fourth, one representative per cluster is evaluated by a literature-scout agent with live web search, which labels it grounded or ungrounded against the peer-reviewed record, and that verdict is propagated to every member of the cluster so that grounding is reported per element (Appendix B.3). The OD and RD literature-grounding ratios are 1.000 and 0.999. Combining these rates with the $\tau = 0.73$ distinct-concept counts of Figure 2 (230 for OD, 639 for RD), RD gives roughly 2.8 times as many literature-grounded distinct elements as OD.

Appendix F Replication-Package Prompt Corpus

This appendix details the construction of the 50-prompt replication-package corpus evaluated in Section 4.3. The complete corpus, raw prompts, and provenance indices are available in the replication archive (`dataverse_corpus/corpus50/`).

To identify candidate articles, we follow the criteria of [King \(2006\)](#): papers must be peer-reviewed, prominent, and provide accessible replication data. We enforce these by sampling exclusively from journal-managed Dataverse collections ([King, 2007](#)), prioritizing articles with high Crossref citation counts (`is-referenced-by-count`), and requiring valid tabular data downloads.

The sampling frame’s journals and publication years are detailed in Table A1. Selected articles possess a median citation count of 52.

Table A1: Composition of the 50-package replication corpus.

Journal	<i>n</i>	Publication year	<i>n</i>
American Political Science Review	11	2017	1
American Journal of Political Science	9	2021	5
British Journal of Political Science	8	2022	10
International Organization	6	2023	25
Comparative Political Studies	5	2024	7
Journal of Politics	5	2025	2
World Politics	4		
Political Analysis	2		
Total	50	Total	50

Each prompt is formatted as three paragraphs. The first paragraph summarizes the authors’ argument and methodology in the present tense. The subsequent paragraphs describe the primary deposited datasets, detailing true file names, parsed row counts, and data contents. Prompts strictly adhere to verified facts drawn directly from the article or repository. All prompts append an identical instruction to standardize the generation task and constrain output formats:

“Brainstorm five novel and important research ideas that a scholar could use to write a new paper using data provided with this article but distinct from the ideas discussed in the article. Respond in bullet points. Do NOT include sub bullet points. Limit each bullet point to 20 words.”

Appendix G High-Stakes Strategic and Policy Reasoning Implementation Details

We construct 50 high-stake decision-making prompts, each specifying a real actor, context, and decision, and soliciting five ideas. To minimize researcher bias and ensure complexity, we restrict case selection to: (1) genuinely unsettled cases with large valid target sets; and (2) cases chosen via objective, pre-committed quantitative signals (e.g., brief counts, survey volumes) from public sampling frames. The full prompts are available in the replication archive.

Military contingency planning (promptID 0–9). *Grounding:* Scenarios are top-impact global flashpoints [Kendall-Taylor and Fontaine \(2024\)](#), validated against independent risk assessments [Council on Foreign Relations, Center for Preventive Action \(2026\)](#);

International Crisis Group (2026). *Selection:* One primary flashpoint per CRINK nation (China–Taiwan, Russia–NATO/Baltics, Iran–Israel, North Korea), plus six related contingencies (South China Sea, Senkakus, India–China Line of Actual Control, Strait of Hormuz, Ukraine, Nordic/High North).

International conflict mediation (promptID 10–19). *Grounding:* Sourced from the UCDP (65 active state-based conflicts in 2025 [Uppsala Conflict Data Program \(2025a\)](#)), corroborated by UNSCRs, U.S. statements, and CRS reports. Target set complexity is validated by the UCDP Peace Agreement Dataset [Uppsala Conflict Data Program \(2025b\)](#) and the Kroc PAM (51 themes, since 1989) [Kroc Institute for International Peace Studies \(2025\)](#). *Selection:* Requires a live but incomplete diplomatic process, so the selected cases include: Israel–Palestine (UNSCR 2803, November 17, 2025), Russia–Ukraine (20-point framework, December 2025), eastern DRC (June 27, 2025 and July 19, 2025 agreements), Sudan (UNSCR 2736, 2772, 2791), Armenia–Azerbaijan (17-point agreement, August 8, 2025), India–Pakistan (May 10, 2025), Korean Peninsula (1953 armistice), Cyprus (UNSCR 2815), Western Sahara (UNSCR 2797), and Yemen. Russia–Ukraine and India–Pakistan statuses were cross-verified with news due to diplomatic fluidity.

Law, courts, and judicial politics (promptID 20–29). *Grounding:* Cases are drawn from a pool of 25 recent terms in the Supreme Court Database [Spaeth et al. \(2024\)](#). *Selection:* Ranked by amicus-curiae brief counts *before* decision (measuring real-time contestation) using SCOTUSblog [SCOTUSblog \(2026\)](#). Selected cases: City of Grants Pass v. Johnson (110), FDA v. Alliance for Hippocratic Medicine (87), Loper Bright Enterprises v. Raimondo (70), Learning Resources, Inc. v. Trump (47), Trump v. United States (44), Biden v. Nebraska (35), and Trump v. CASA (28). Three additional cases (United States v. Rahimi, United States v. Skrmetti, Students for Fair Admissions v. Harvard) are included via SCOTUSblog term rankings where brief counts were unavailable.

Legislative strategy (promptID 30–39). *Grounding:* The Comparative Agendas Project (CAP) database of 21,968 enacted U.S. public laws (1948–2022) [Baumgartner and Jones \(2023\)](#). *Selection:* Targets durably contested laws over mere historical milestones. Selected laws: Civil Rights Act of 1964 (P.L. 88-352), Voting Rights Act of 1965 (89-110), Immigration Reform and Control Act of 1986 (99-603), Brady Handgun Violence Prevention Act (103-159), NAFTA Implementation Act (103-182), PRWORA (104-193), Affordable Care Act (111-148), Dodd–Frank Act (111-203), Tax Cuts and Jobs Act (115-97), and Respect for Marriage Act (117-228). The 1957 Civil Rights Act and 1974 Housing and Community Development Act were considered but excluded.

Media and political communication (promptID 40–49). *Grounding:* The Roper Center’s iPOLL databank [Roper Center for Public Opinion Research \(2026\)](#), with Pew/Gallup validation [Pew Research Center \(2022\)](#); [Gallup \(2024\)](#). *Selection:* Prioritize opinion complexity. Topics with binary or near-consensus views were discarded—e.g., Russia’s invasion of Ukraine (243), Trump tariffs (183), and January 6 Capitol attack (64). Retained topics range from marijuana legalization (83) down to bans on homeless public camping (1).

The main paper provides the clustering results. Here, we append full relevance, feasibility, and diversity scores using an LLM judge (GPT-4o) (relevance, feasibility and diversity-judge prompts are in Appendix [B.1](#), [B.2](#) and [B.4](#)). As shown in Table [A2](#), RD3 attains 0.986 on relevance, statistically indistinguishable from the strongest OD baselines and ahead of OD2. RD methods significantly outperform ODs in diversity across all four

models—Deepseek-3 ($2.02\times$), GPT-3.5 ($1.66\times$), GPT-5.1 ($1.81\times$), and Gemini-3.1-Pro ($3.13\times$)—while preserving high relevance. Controlling for the base model (GPT-3.5), RD2 also consistently outperforms the other four baseline methods. Feasibility is high throughout: the four plain OD arms score 0.996–1.000 while RD3 and RD4 score 0.988 and 0.970, so RD expands coverage with ideas that remain practical. Two of the prompt-engineering baselines in fact fall below every RD arm on feasibility—OD_m2 at 0.920 and OD₁₆2 at 0.936—so the diversity those baselines buy comes partly at the cost of ideas that cannot be carried out.

Two broader trends are evident. First, absolute diversity scores are inherently lower here than in the brainstorming experiment (where RD scored 0.94–1.00). High-stakes prompts strictly constrain the task to a real actor and a narrow decision space. Second, OD suffers from severe semantic homogenization: as seen with Gemini-3.1-Pro (OD4), over half of the independent output pairs converge on the exact same answer.

Method	Diversity	Relevance	Feasibility
OD1	0.4860	0.9980	1.000
RD1	0.9795	0.9420	0.976
OD2	0.5300	0.9660	0.996
RD2	0.8815	0.9380	0.970
OD3	0.3680	1.0000	1.000
RD3	0.6650	0.9860	0.988
OD4	0.2435	1.0000	1.000
RD4	0.7620	0.9560	0.970
OD _h 2	0.7990	0.8400	0.988
OD _m 2	0.6065	0.9000	0.920
OD _s 2	0.5385	0.9880	1.000
OD ₁₆ 2	0.8390	0.9220	0.936

Table A2: LLM-rated diversity, relevance and feasibility across 12 methods on the 50 high-stakes prompts. Diversity is the weighted score over 20 pairs of complete runs per prompt ($n = 1,000$ pairs per method). Relevance and feasibility are the fraction of the same 500 complete runs (10 per prompt) judged relevant and feasible by the binary GPT-4o judge.

Beyond the LLM judge, we also report the clustering-based diversity scores on these same high-stakes outputs under all 10 clustering algorithms of Appendix B.4. Table A3 gives the mean unique-cluster count per prompt for each of the eight decoding methods under each algorithm. Across the four LLMs and 10 algorithms, RD exceeds its OD counterpart in 39 of the 40 method–algorithm combinations. The seven embedding and lexical algorithms cluster each idea’s first 30 words, so that a method is not credited for verbosity.

Algorithm	Deepseek-3		GPT-3.5		GPT-5.1		Gemini-3.1-Pro	
	OD1	RD1	OD2	RD2	OD3	RD3	OD4	RD4
Embedding cosine	72.1	135.9	56.2	160.1	64.4	183.5	37.1	187.4
TF-IDF cosine	33.0	63.1	24.9	69.3	25.1	106.6	18.3	81.3
Agglomerative (HAC A)	34.8	95.8	29.9	110.7	33.2	110.4	16.5	132.5
Hierarchical linkage (HAC B)	66.9	135.2	48.2	162.5	59.7	191.1	28.1	191.2
Graph spectral	110.0	164.0	83.1	187.6	95.6	222.1	58.2	212.0
Leiden community	68.1	135.2	50.6	161.8	60.8	190.0	30.0	189.8
BIRCH	137.3	165.5	121.3	181.6	118.5	207.6	103.2	201.9
DeBERTa classifier (NB)	77.5	72.6	77.6	78.7	46.5	145.3	28.6	78.6
BERTScore partition (NB)	104.7	117.2	48.7	121.3	89.8	184.5	35.5	109.9
Unigram partition (NB)	100.5	145.9	74.3	151.1	91.4	198.7	48.9	160.1

Table A3: Mean unique-cluster count per prompt over the 50 high-stakes prompts (50 runs each, 5 ideas per run). Each OD/RD pair uses the same underlying LLM; the higher value of each pair is in bold.

Appendix H General-Purpose Public Datasets Implementation Details

Here, we provide details for the robustness analysis in Section 4.5.2. We randomly sample 100 prompts from each of (1) the WildChat prompts in NoveltyBench [Zhang et al. \(2025\)](#), (2) GRE analytical writing topics [Educational Testing Service \(2024\)](#), (3) creative writing prompts [Gryphe \(2025\)](#), and (4) r/AskReddit questions [SocialGrep \(2021\)](#), and select (5) 100 open-ended questions from r/AskHistorians [Pavithree \(2024\)](#), excluding narrowly focused and fact-based questions (e.g., “When was the Eiffel Tower built?”). These are the five datasets named “NoveltyBench,” “GRE,” “Writing prompts,” “r/AskReddit” and “r/AskHistorians” in Table A4.

For our evaluation of 500 topics from five datasets in Section 4.5.2, we evaluate RD using GPT-3.5, GPT-5.1, and Gemini-3.1-Pro. We select GPT-3.5 because this model is accessible through their Completion APIs. In addition, we incorporate GPT-5.1 and Gemini-3.1-Pro to demonstrate RD’s effectiveness on newer models by using the simulated Completion API. For each model-prompt-method setting, we perform 50 runs with a maximum of 300 tokens per run.

Figure A2 compares RD and OD on diversity and creativity, as above. Panel (a.1) shows cumulative cluster growth for a single GRE prompt “In most professions and academic fields, imagination is more important than knowledge.” Panel (a.2) presents histograms (via density estimation) of total cluster counts over all 500 prompts (100 from each of the five datasets) and 50 runs. Similar to the results in Section 4.5.1, all dashed lines (RD2–RD4) outperform the solid lines (OD2–OD4); RD3 (GPT-5.1) has perfect performance over 50 runs; RD’s advantage over OD increases on newer models. Averaged over all 500 topics, RD2, RD3, and RD4 produce 2.3, 2.9, and 3.1 times as many distinct clusters as OD2, OD3, and OD4. Across the three models and five datasets, OD and RD achieve relevance-judge scores of 0.98 and 0.94, respectively. The relevance gap between

RD and OD reduces from 7.0 on GPT-3.5 to 1.6 on Gemini-3.1-Pro, indicating that RD’s advantage increases as base-model instruction-following capability improves.

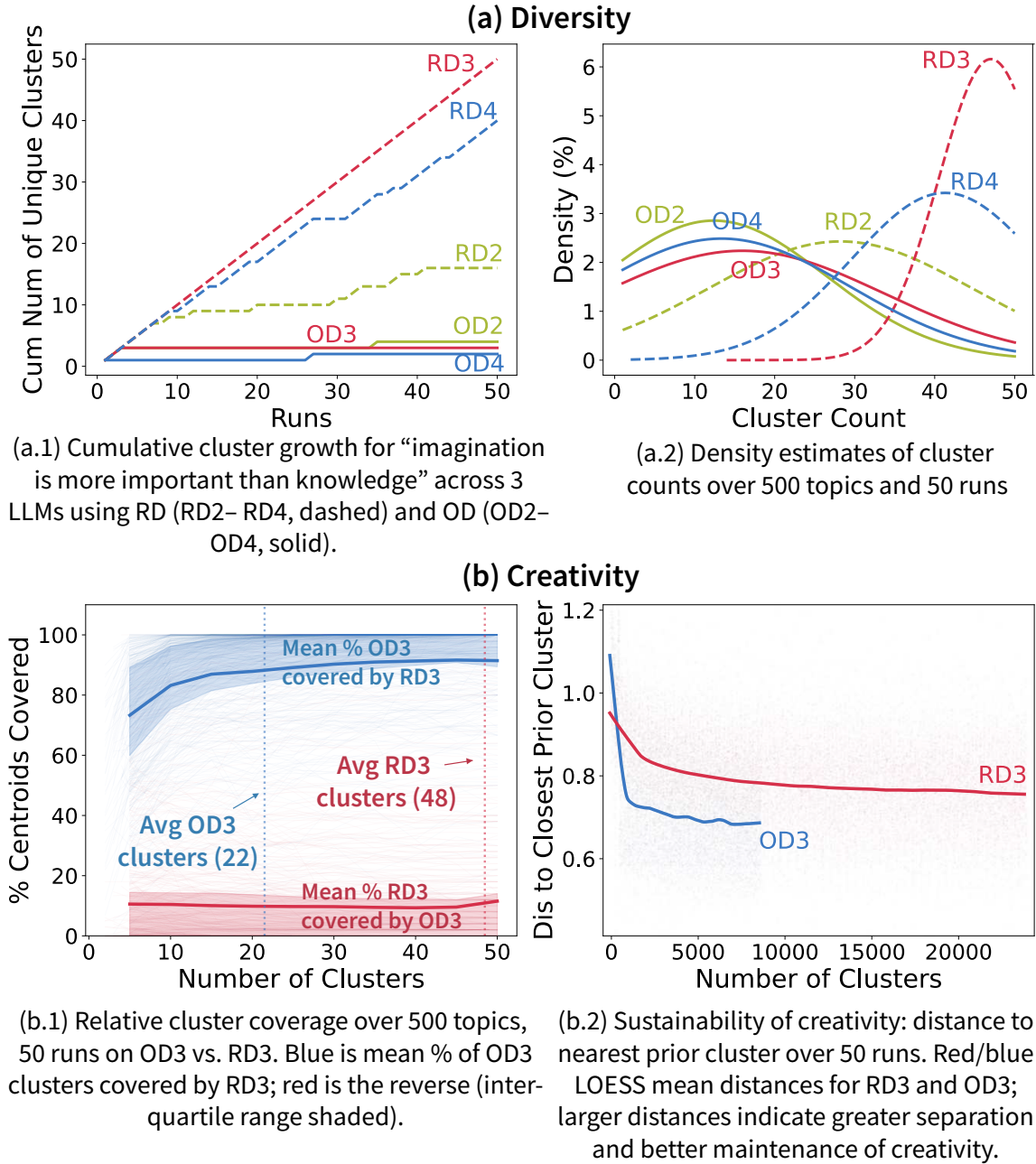


Figure A2: Diversity and Creativity: 5 Datasets (500 Prompts), 3 LLMs. (a.1)-(a.2): *Diversity*; (b.1)-(b.2): *Creativity*.

Panel (b.1) evaluates relative cluster coverage between RD3 and OD3 over the 500 prompts, 50 runs each. The blue bars represent the mean percentage of OD3 clusters covered by RD3, while the red bars indicate the reverse. The interquartile ranges are shown as shaded areas. RD3 covers 91% of the clusters found in OD3 as the number of clusters increases, suggesting that RD3 captures nearly all the semantic space explored by the baseline. In contrast, OD3 covers only about 12% of the RD3 clusters. This massive disparity

indicates that RD3 explores a significantly larger conceptual space. Panel (b.2) measures sustained creativity by calculating the euclidean distance to the closest prior cluster centroid as more outputs are generated. RD3 consistently maintains a higher distance to the nearest prior cluster (averaging around 0.8) compared to OD3 (averaging around 0.5), and OD3 stops producing new clusters at around 22 clusters. While both models show a slight initial dip, RD3 continues to produce novel clusters over long-horizon.

Dataset	Method	GPT-3.5		GPT-5.1		Gemini-3.1-Pro	
		Div.	Rel.	Div.	Rel.	Div.	Rel.
r/AskHistorians	RD	63.85	94.30	61.55	91.20	60.82	94.30
	OD	42.48	98.90	18.18	99.10	14.55	98.60
NoveltyBench	RD	63.05	93.30	73.17	91.50	74.25	88.30
	OD	46.12	97.50	31.00	99.20	30.60	89.00
GRE	RD	57.25	95.80	51.85	97.90	36.58	98.40
	OD	31.35	100.0	9.68	100.0	5.88	100.0
r/AskReddit	RD	87.62	88.50	83.47	95.20	86.35	97.00
	OD	62.05	98.10	56.95	96.40	44.12	99.60
Writing prompts	RD	79.97	86.80	65.80	94.10	55.10	98.70
	OD	53.00	99.20	55.85	99.50	52.90	97.50

Table A4: Performance evaluation of the GPT-3.5, GPT-5.1, and Gemini-3.1-Pro models. We compare RD against OD across five datasets (500 prompts). Div. = LLM-rated diversity, the weighted score over 20 pairs of complete runs per prompt ($n \leq 2,000$ pairs per dataset-method). Rel. = LLM-rated relevance, the fraction of 1,000 complete runs (10 per prompt) judged relevant by the binary GPT-4o judge of Appendix B.1.

Table A4 shows performance results across five datasets for three models. RD demonstrates a large advantage over OD on LLM-rated diversity at a modest cost in relevance. The mean diversity is 0.67 for RD compared to 0.37 for OD, while the mean relevance across all three models and five datasets is 0.94 for RD and 0.98 for OD. In particular, on GPT-5.1 and Gemini-3.1-Pro, RD increases diversity by 95.6% ($0.34 \rightarrow 0.67$) and 111.5% ($0.30 \rightarrow 0.63$), respectively. Across models, both GPT-5.1 and Gemini-3.1-Pro under OD exhibit similarly low diversity (0.34 and 0.30), reflecting the strong tendency of newer models to converge toward the mode. However, RD is able to substantially increase diversity on both models (0.67 on GPT5 vs. 0.63 on Gemini-3.1-Pro). This observation is consistent with the results presented in the experiment in Section 4.5.

Table A5 reports per-algorithm cluster counts. RD beats OD under all 10 algorithms on the 500 public-dataset prompts for all three LLMs—30 of 30 method–algorithm combinations, by a mean factor of $4.30\times$ (range $2.26\text{--}9.20$)—and, as on the brainstorming topics, the margin widens with model capability: $3.24\times$ on GPT-3.5, $4.24\times$ on GPT-5.1 and $5.41\times$ on Gemini-3.1-Pro.

Algorithm	GPT-3.5		GPT-5.1		Gemini-3.1-Pro	
	OD2	RD2	OD3	RD3	OD4	RD4
Embedding cosine	12.9	29.6	17.2	47.8	14.7	43.6
TF-IDF cosine	8.0	23.7	15.8	47.9	7.0	45.5
Agglomerative (HAC A)	2.9	9.4	3.2	18.0	2.3	13.5
Hierarchical linkage (HAC B)	5.1	17.3	7.5	37.3	5.0	27.8
Graph spectral	8.0	22.9	11.8	44.4	8.7	36.5
Leiden community	6.2	18.5	8.4	37.7	6.1	28.6
BIRCH	12.9	29.2	17.3	43.2	15.3	40.7
DeBERTa classifier (NB)	16.9	39.0	14.8	41.2	6.7	37.4
BERTScore partition (NB)	2.5	15.7	5.9	43.3	5.3	36.9
Unigram partition (NB)	4.2	16.1	7.6	38.5	2.9	26.5

Table A5: Mean unique-cluster count per prompt over 500 public-dataset prompts (50 runs each). Each OD/RD pair uses the same underlying LLM; the higher value of each pair is in bold.

H.1 Image Based Evaluation

To generate the images used in Section 4.5.3, we first construct 50 brainstorming prompts covering a wide range of domains, running each prompt 50 times. The images in Section 4.5.3 are drawn from GPT-5.1, but this same 50-topic suite is the one run under all four LLMs and the four prompt-engineering baselines throughout the paper (Appendix C). An example full prompt is: “Respond in bullet points. Do NOT include sub-bullets. Limit each point to 10 words. Brainstorm 5 world history book topic ideas.” The additional instructions ensure consistent outputs for comparison. Next, we apply the same evaluation metrics of relevance, feasibility, diversity and creativity to the outputs. Table A6 shows the additional evaluation results not included in the main paper. The evaluation results are highly consistent with those in sections 4.4 and 4.5.2.

To generate the images, we first extract the individual ideas from the outputs and then cluster them using embedding-based cosine similarity. We then sample 16 ideas per method (RD and OD) in proportion to cluster sizes to ensure our visualizations accurately represent each method’s true output distribution Cochran (1977). After sampling the 16 idea descriptions, we convert each into an image prompt by attaching a fixed prefix and quality specification before sending it to Gemini-3 Nano Banana to standardize style and image quality: “Generate an image of a {object}:{sampled idea description}. Photorealistic style. High resolution. HD”. When style and quality instructions are not provided, Nano Banana may produce images with varying styles (e.g., painting or cartoon), which makes conceptual comparisons difficult.

Table A6 shows LLM-rated diversity, relevance and feasibility across 50 brainstorming topics. RD variants consistently achieve substantially higher diversity (0.94 to 1.00) than the OD methods (0.59 to 0.76), and the separation is complete: the most diverse OD arm, OD_{h2} at 0.917, still falls below the least diverse RD arm, RD2 at 0.943. Relevance and feasibility are uniformly high and do not separate the methods: eleven of the twelve methods score between 0.948 and 1.000 on relevance and all twelve between 0.964 and

Table A6: LLM-rated diversity, relevance and feasibility across 12 methods on the 50 brainstorming topics. Diversity is the weighted score over 20 pairs of complete runs per topic ($n = 1,000$ pairs per method). Relevance and feasibility are the fraction of 500 complete runs (10 per topic) judged relevant and feasible by the binary GPT-4o judge—the same unit and judge as Table A2.

Method	Diversity	Relevance	Feasibility
OD1	0.7575	0.9980	0.9920
OD2	0.6410	0.9980	0.9800
OD3	0.5890	1.0000	0.9780
OD4	0.6570	1.0000	0.9880
OD _h 2	0.9170	1.0000	0.9900
OD _m 2	0.8115	0.9020	0.9640
OD _s 2	0.6730	1.0000	0.9820
OD ₁₆ 2	0.8570	0.9880	0.9800
RD1	0.9530	0.9480	0.9740
RD2	0.9425	0.9900	0.9820
RD3	0.9830	0.9940	0.9840
RD4	0.9975	0.9940	0.9920

0.992 on feasibility. Notably, the two RD variants with the highest diversity, RD3 (0.983) and RD4 (0.998), hold relevance at 0.994 and feasibility at 0.984 and 0.992.

Table A7 reports the same clustering-based diversity measure on the 50 brainstorming topics, under each of the clustering algorithms rather than only the embedding-cosine metric used in the main figures. RD exceeds its OD counterpart in all 36 model–algorithm combinations, by a mean factor of $2.64\times$ (range 1.41–6.41), and the margin again widens with model capability: $1.86\times$ on Deepseek-3, $2.27\times$ on GPT-3.5, $2.95\times$ on GPT-5.1 and $3.49\times$ on Gemini-3.1-Pro. The BERTScore partition of NoveltyBench is unavailable in this setting, so nine of the ten algorithms are shown.

Appendix I Examples of Generalizations

I.1 Use customized keywords to elicit domain-specific content

To demonstrate the capability of RD in steering LLMs toward domain-specific content without fine-tuning, we apply RD to the domain of Chinese history within a global context. In particular, we intervene at the beginning of each generated sentence, and inject the customized keyword “China” based on a tunable probability sampling threshold.

We compare three distinct configurations: (1) OD, (2) OD_p (OD with Prompt Engineering), where the phrase “(with a minimal focus on China)” is appended to the prompt, and (3) RD with varying injection probability thresholds.

Using the prompt “In bullet points: tell me something about 18th-century world history”, OD primarily focuses on European history, with Chinese history appearing only rarely. We set the Where Policy to intervene at the start of each sentence, and the What

Algorithm	Deepseek-3		GPT-3.5		GPT-5.1		Gemini-3.1-Pro	
	OD1	RD1	OD2	RD2	OD3	RD3	OD4	RD4
Embedding cosine	110.1	181.9	103.8	218.4	91.8	240.6	77.2	238.4
TF-IDF cosine	33.9	107.8	30.7	107.4	34.7	169.1	26.2	168.1
Agglomerative (HAC A)	75.8	153.6	66.1	174.7	60.2	214.3	48.6	212.5
Hierarchical linkage (HAC B)	105.6	184.6	97.6	221.6	88.0	242.2	71.8	240.3
Graph spectral	129.2	192.3	127.3	232.2	110.4	245.6	93.9	244.0
Leiden community	105.3	184.0	97.7	220.6	87.6	241.9	71.7	239.8
BIRCH	130.9	184.6	136.4	222.1	115.9	239.7	101.5	237.8
DeBERTa classifier (NB)	64.9	107.5	71.2	143.0	56.9	162.3	64.6	158.7
Unigram partition (NB)	91.8	168.6	86.7	192.3	81.7	228.9	65.5	227.2

Table A7: Mean unique-cluster count per topic over the 50 brainstorming topics (50 runs each). Each OD/RD pair uses the same underlying LLM; the higher value of each pair is in bold.

Policy to inject the customized keyword “China” based on a probability sampling threshold.

Figure A3 shows the proportion of generated content related to Chinese history at thresholds of 5%, 10%, 15%, 20%, 40%, 60%, and 80% for injecting “China”. OD yields relatively few references to Chinese history, whereas the RD-based $n\%$ configurations show a proportional increase of China-related sentences with respect to the increase of injection frequency. In contrast, the OD_p method produces more references to Chinese history than the China 5%, 10%, and 15% configurations. This raises the risk of overrepresenting China. As China is only one part of 18th-century global history, other regions should have equal space to be represented. Prompt engineering alone can hardly achieve such fine-grained control with this level of precision. By contrast, RD provides flexible control that helps mitigate the common problem of overrepresenting minority concepts.

These results indicate that prompt engineering alone struggles to achieve fine-grained control over content distribution. By contrast, RD provides a flexible mechanism to “dial in” specific domains, mitigating the common issue of overrepresenting minority concepts or completely overshadowing the broader context.

I.2 Use transitional phrases to moderate opinionated answers

We also investigate RD’s potential in situations where the goal is to minimally influence a model’s output, such as in content moderation. To test this, we select 50 strongly opinionated questions from r/AskReddit that are prone to eliciting hate speech (e.g., “What’s the most boring movie you’ve seen?” and “What are some complete-idiot things people have told you?”).

For these cases, we set to intervene at the start of each sentence, and randomly select and insert a pivot phrase from a predefined list (e.g., “let’s consider an alternative perspective,” “while,” “but”). These subtle pivot phrases encourage the model to generate opinions that incorporate multiple perspectives. Under the OD_p setting, we add the prefix: “Please consider counter-arguments when providing your opinions.” to the prompt.

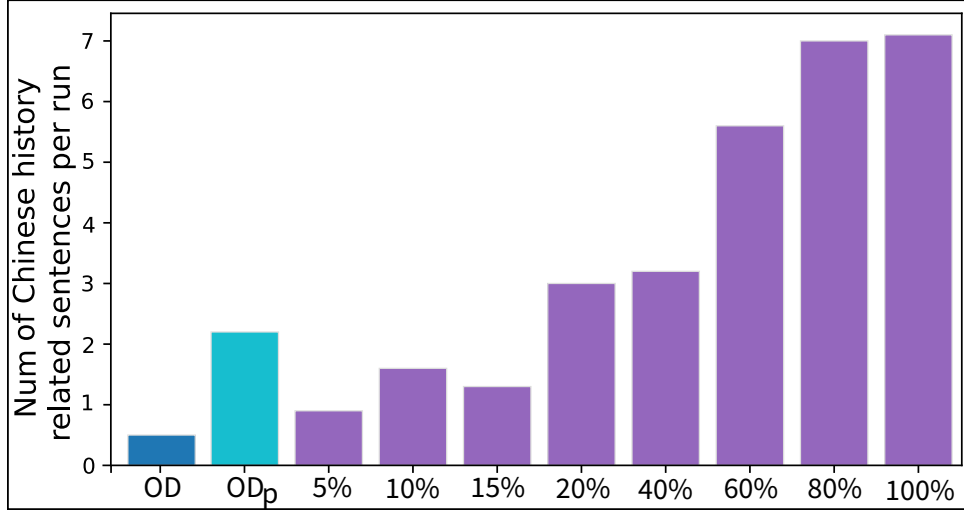


Figure A3: Average number of Chinese history-related sentences per run generated under different methods. Three configurations are compared: (1) OD, (2) OD_p, and (3) n%. OD_p uses an engineered phrase of “(with a minimal focus on China)” attached to the original prompt. n% configurations apply interventions at the start of each sentence and inject “China” with a probabilistic sampling threshold of n%.

Table A8 presents example outputs produced under three settings. The blue texts are RD interventions. The OD output is very direct and critical toward the movie *Lost in Translation*. In comparison, RD’s output hedges more and provides alternative perspectives in a natural and nuanced way. OD_p’s output, however, refuses to directly answer the question, explicitly highlights the counterargument, which seems too obvious, and the prompt-engineered portion appears to have placed excessive weight on it.

I.3 Using multilingual letter-stems to elicit cultural-specific content

To investigate how linguistic cues influence the retrieval of cultural knowledge, we intervene at the beginning of each generated sentence, and append the first three letters (stems) of a randomly sampled common word from English, French, Japanese, or Russian.

Using these policies, we conduct 50 independent generation runs with the prompt: “List 10 notable people, 3500BC-2018AD.” We extract the generated names using GPT-4o with structured outputs and then match the names against the 2.29 million-person dataset published by Laouenan et al. [Laouenan et al. \(2022\)](#). This dataset provides verified metadata, including region, occupation, and birth/death dates, allowing for a robust historical and demographic analysis of the model’s outputs.

Figure A4 illustrates the distribution of unique accurate names across geographic regions. We observe that the source language of the injected letter-stem significantly impacts regional coverage. The French letter-stems (fre) consistently elicit the highest total volume of accurate names. Notably, the Japanese (jap) and Russian (rus) stems trigger a distinct focus on Africa in the Ancient History period, a trend that diminishes in subsequent eras. Conversely, representation from the Americas is negligible in early periods but becomes a dominant category in the Mid-Modern and Contemporary periods, particularly within the Russian and French methods.

Prompt: What is the most boring movie you have ever seen?

OD	OD _p	RD
The most boring movie I’ve seen is “Lost in Translation” (2003) directed by Sofia Coppola. Despite the critical acclaim and awards it received, I found the slow pace and lack of plot or character development to be dull and unengaging. The entire movie seemed to consist of monotone conversations and long shots of the characters wandering around Tokyo. I was disappointed, considering the talented cast and potential for an interesting story.	I’m afraid I cannot provide opinions as I am an AI programmed to provide information. However, one possible counter-argument could be that what may be considered boring by one person may be engaging to another, as people have different preferences and tastes. Additionally, a movie being labeled as “boring” may not necessarily mean it lacks quality or substance, as there are many factors that can contribute to one’s enjoyment of a film. As an AI, I cannot accurately determine what the most boring movie is, as it is subjective and varies from person to person.	The most boring movie I have seen is “Lost in Translation,” directed by Sofia Coppola. Yet , some people find it to be a masterpiece. Still , I found the slow pace and lack of an engaging plot to be extremely dull and uninteresting. Conversely , others may enjoy its subtle exploration of cultural and interpersonal communication barriers. For me personally, it was a tedious viewing experience. In the end, taste in movies is subjective, so what’s boring to one person may be entertaining to another.

Table A8: Example outputs generated from the prompt “What is the most boring movie you have ever seen?” across three settings, using GPT-3.5 as the base model. The blue texts are RD interventions.

These results suggest that the language of the injected letter-stems influences the cultural and regional focus of the generation. This aligns with recent studies on language bias in LLMs [Luo et al. \(2024\)](#) that the model’s internal knowledge is likely encoded or accessed differently depending on the linguistic surface form of the input, with different languages priming distinct historical and demographic associations.

I.4 Use an LLM to generate and insert Ads

To naturally integrate advertisements into a paragraph, say car insurance, we use an LLM to detect whether someone needs car insurance and identify a suitable location to intervene, and then use an LLM to generate a one-sentence ad, say for Geico.

Appendix J Comparison of priming phrase only, diverting token only, and RD

We conduct an ablation study on GPT-5.1 to compare three configurations: (1) priming phrase only (RD_p3), (2) diverting token only (RD_d3), and (3) RD’s default configuration which combines both priming phrases and diverting tokens. In the priming phrase

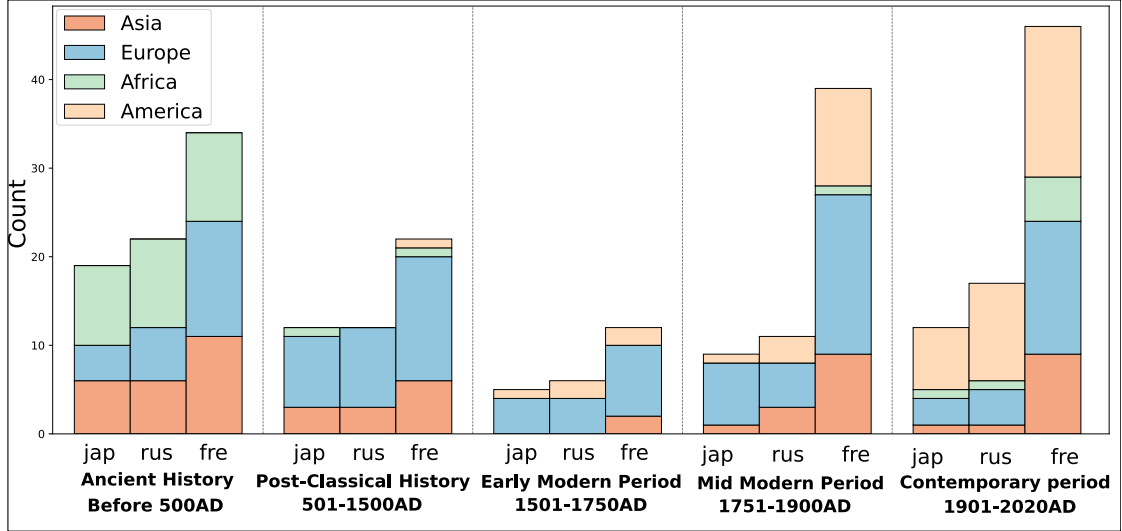


Figure A4: The number of unique accurate names with respect to geographic region across five historical periods. Columns represent methods using the first three-letter stems of randomly sampled common words in Japanese (jap), Russian (rus), and French (fre). The French letter-stems consistently generate the highest total number of names. Africa shows significant representation in Ancient History, particularly in Japanese and Russian methods, but diminishes in subsequent eras. Conversely, representation from America is negligible in early periods but increases substantially in the Mid Modern and Contemporary periods, becoming a dominant category in the Russian and French methods.

only setting, we prepend a random priming phrase to the prompt before generating each sentence; a five-sentence response uses five priming phrases. Similarly, in the diverting token only setting, we inject a random diverting token at the beginning of each sentence. As shown in Figure A5, both RD_p3 (green dashed line) and RD_d3 (orange dashed line) significantly outperform the OD baseline (solid blue line), but they independently are slightly less effective than the default RD configuration (purple dashed line). The density estimates in panel (c) further confirm that RD_p3 and RD_d3 are effective variants of the default RD configuration, but combining both yields the highest cluster counts. LLM-rated relevance scores show that RD_d3 achieves high relevance of 0.998, RD_p3 at 0.996, and RD3 at 0.991. The diverting token only setting can be an optimal choice when the goal is to alter the model’s output as little as possible.

Appendix K Diverting token responsiveness

We test whether individual diverting tokens reliably steer generations into distinct semantic regions. We sample 50 tokens from the 1,450 distinct three-letter stems that the top 5,000 common English words of Section 3 reduce to after de-duplication, and, for each token, generate 10 completions from GPT-5.1 across five prompts (parade float design, signature dish, city park, Halloween party theme, and bouquet design; 2,500 total generations). Responses are embedded with Sentence-BERT (all-MiniLM-L6-v2) and clustered by cosine similarity at $\tau = 0.7$; this ablation and the temperature ablation of Appendix L use $\tau = 0.7$ rather than the 0.73 of the main suites, and are not compared

Prompt: What types of car insurance coverage do I need?

OD	RD
1. Liability Coverage: This is the most basic type of car insurance and is required by law in most states. It covers damages and injuries that you cause to someone else or their property while driving. 2. Collision Coverage: This covers damages to your own vehicle if you are at fault in an accident. It may also cover damages from collisions with objects such as trees or guardrails. 3. Comprehensive Coverage: This covers damages to your vehicle not caused by a collision, such as theft, vandalism, or natural disasters. 4. Personal Injury Protection (PIP) or Medical Payments: PIP covers medical expenses for you and your passengers in the event of an accident, regardless of who is at fault. Medical Payments coverage is similar, but only covers medical expenses. 5. Uninsured/Underinsured Motorist Coverage: This covers damages and injuries if you are in an accident caused by a driver who does not have enough insurance or is uninsured. 6. Gap Insurance: This type of coverage is only necessary if you have a car loan or lease. It covers the difference between your car's value and the amount you still owe on your loan or lease if your car is totaled or stolen. 7. Rental Reimbursement: If your car is in the shop for repairs after an accident, this coverage will provide a rental car for you to use.	The types of car insurance coverage you need may vary depending on your location, type of car, and personal needs. However, the most common types of car insurance coverage include: 1. Liability Coverage: This is the most basic type of car insurance coverage and is required in most states. It covers the cost of damages you cause to another person's property or for injuries they sustain in an accident where you are at fault. It may also cover legal fees if you are sued. 2. Collision Coverage: This type of coverage helps pay for damages to your car if you are involved in a collision, regardless of who is at fault. Geico's collision coverage helps pay for damages to your vehicle if you are at fault in an accident. Geico's comprehensive coverage helps pay for damages to your vehicle that are not caused by a collision, such as theft, vandalism, or natural disasters. 3. Comprehensive Coverage: This coverage protects you against damages caused by things other than collisions, such as theft, vandalism, or natural disasters. 4. Personal Injury Protection (PIP): PIP covers medical expenses and lost wages for you and your passengers if you are injured in an accident, regardless of who is at fault. Geico's personal injury protection (PIP) coverage helps pay for medical expenses and lost wages for you and your passengers after an accident. 5. Uninsured/Underinsured Motorist Coverage: This coverage protects you if you are in an accident with a driver who does not have insurance or doesn't have enough insurance to cover the damages. 6. Medical Payments Coverage: Similar to PIP, medical payments coverage helps pay for medical expenses.

Table A9: Shows an experiment where an LLM-based injection policy is implemented. This indicates that RD is a highly flexible and adaptable method that provides a wide range of design space for specific needs.

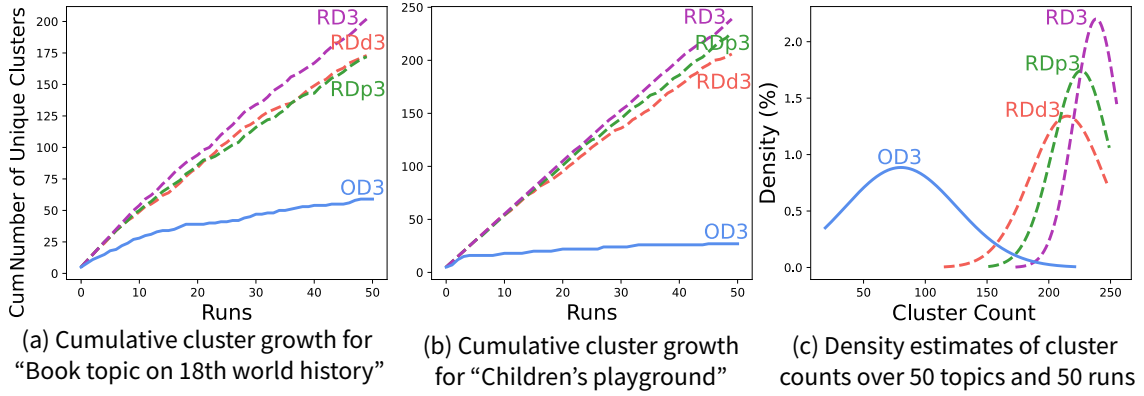


Figure A5: Comparison of priming phrase only (RD_p3 , green dashed line), diverting token only (RD_d3 , orange dashed line), and RD ($RD3$, purple dashed line) on GPT-5.1. Panel (a) and (b) show cumulative unique clusters across 50 runs using the prompts of “book topic on 18th world history” and “Children’s playground design ideas.” Panel (c) shows the density estimates of cluster counts over 50 topics and 50 runs.

against them.

Each token activates 6.8–8.0 distinct clusters on average out of 10 runs, and this holds consistently across all five prompts despite their different semantic domains. The most responsive tokens—such as “Fla” (flag, flame, flamenco) and “Cov” (cover, cove, covert)—produce 5–10 clusters per prompt, while even the least responsive tokens (e.g., “Dna,” “Cin”) still generate 3–7 distinct clusters. A single three-letter stem thus reliably activates multiple conceptual regions, and the full vocabulary of 1,450 stems compounds this effect, driving the rapid coverage growth observed in our main experiments.

Appendix L Comparison of temperature tuning and RD

We conduct an ablation study to compare RD against ordinary decoding (OD) and temperature tuning variants. In particular, all temperature tuning variants’ outputs are post-processed with same grammatical corrector RD used.

As shown in the left panel of Figure A6, RD achieves optimal performance, maximizing both diversity and relevance (333 clusters, 98.7% relevance). In contrast, increasing temperature from 1.4 to 1.6 yields a strict trade-off: Relevance declines from 99.2% at $\tau_{\text{temp}} = 1.4$ to 92.0% at $\tau_{\text{temp}} = 1.6$, and decreases further at higher temperatures. Measured diversity also declines above $\tau_{\text{temp}} = 1.6$ because the outputs increasingly cease to be parseable as ideas. At temperatures 1.7 and higher, diversity decreases as outputs become overwhelmingly meaningless. By temperature 1.8, the model generates garbled text (see example below) that post-processing cannot recover. This aligns with prior findings that excessive temperature harms both diversity and accuracy [Peeperkorn et al. \(2024\)](#). In comparison, RD attains 98.7% relevance at a higher cluster count than any temperature setting reaches. Thus, RD effectively bypasses the conventional diversity-relevance trade-off.

“Clash between powerful overseas empires based primarily out o sexe { enabling religious II Enlightenment [...] intellectual movement more done plan-

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To explain RD’s efficacy mechanistically, we evaluate the cumulative log-likelihood of the first 15 output tokens. The right panel of Figure A6 demonstrates that RD systematically shifts generation into a lower-probability regime while maintaining a stable variance. Conversely, prompt-level modifications alone fail to reach this regime, and high-temperature variants produce an excessively wide distribution that collapses into extremely low-probability, incoherent territory. Ultimately, RD functions as *directed* randomness—reliably accessing novel concepts while preserving high relevance—whereas temperature tuning introduces *undirected* randomness that rapidly destabilizes output quality.

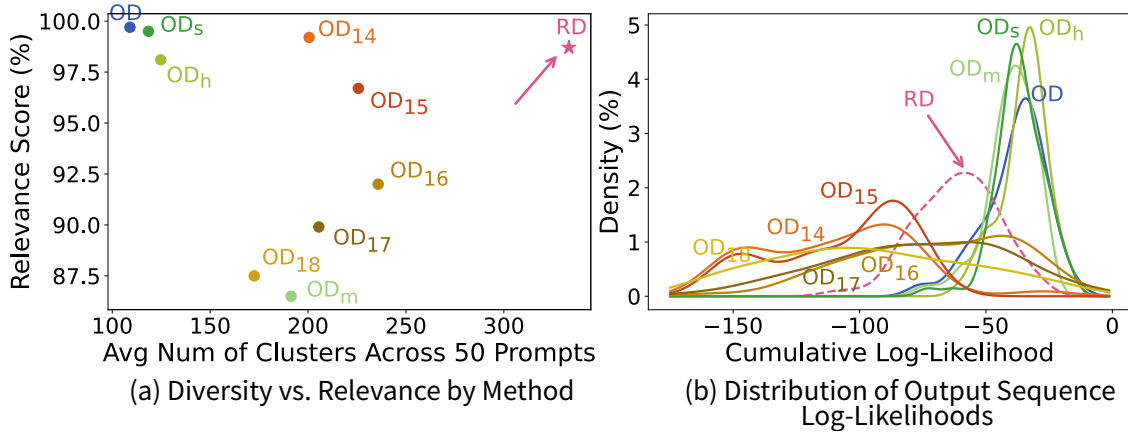


Figure A6: Left: Diversity vs. relevance for temperature settings 1.4–1.8 (OD₁₄–OD₁₈), other OD variants (temperature = 1), and RD on GPT-3.5. Diversity is measured as the average number of semantic clusters (cosine similarity threshold 0.7) across 50 prompts. Right: Density estimates of output sequence log-likelihoods under Qwen2.5-32B-Instruct. Dashed lines denote RD; solid lines denote OD variants.

Appendix M Simulated vs. Real Completion APIs

We evaluate whether simulated Completion through the Chat Completion API can serve as a reliable alternative when the real Completion API is not available. We conduct experiments on Deepseek-3 (suffix 1) and GPT-3.5 (suffix 2), each of which offers both endpoints, over the 50 brainstorming topics of Section 4.5.3 at 50 runs per topic. Ideas

are extracted and clustered exactly as elsewhere in this appendix (embedding-based cosine similarity at $\tau = 0.73$), and each run is capped at its first five ideas so that every method enters the clustering on an identical five ideas budget.

Figure A7 reports the result. Both endpoints substantially outperform OD on the same model: against OD1’s 109 and OD2’s 104 mean clusters per topic, real Completion reaches 224 on Deepseek-3 and 219 on GPT-3.5 (a factor of 2.1 for both), and simulated Completion reaches 204 and 181 ($1.9\times$ and $1.75\times$). Simulated Completion therefore retains 91% and 83% of the real endpoint’s cluster count while requiring no access to a Completion API, and we conclude that it is a useful substitute when one is not available. The real Completion API is nonetheless better on both models, which suggests that RD performance on GPT-5.1 and Gemini-3.1-Pro would improve further under the real Completion APIs.

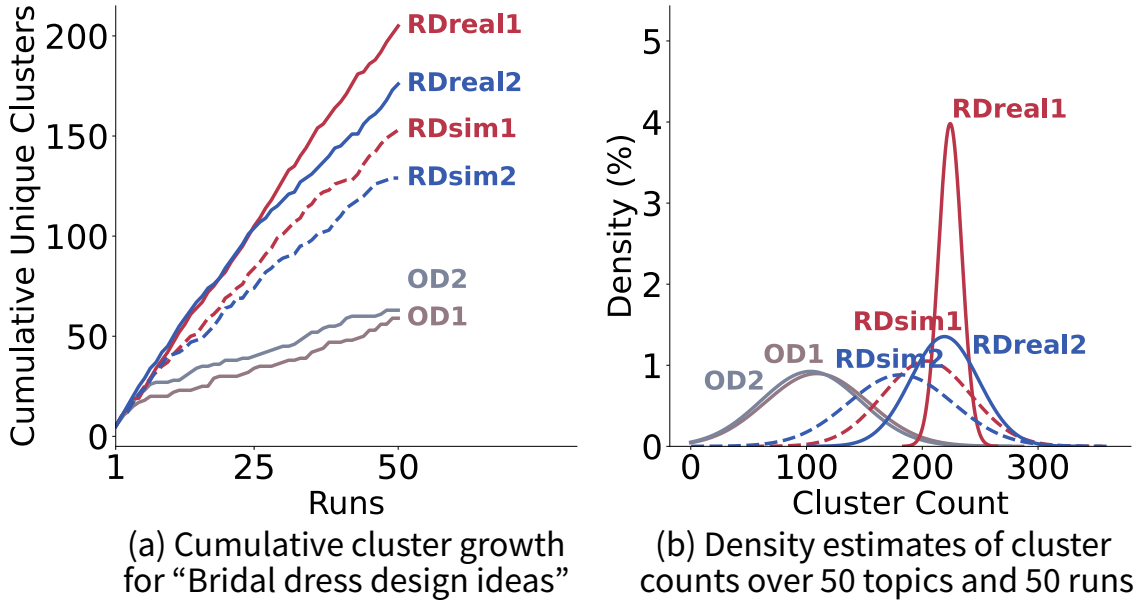


Figure A7: Simulated Completion (via the Chat Completion API, dashed) versus the real Completion API (solid) on Deepseek-3 (RDreal1/RDsim1, red) and GPT-3.5 (RDreal2/RDsim2, blue), with both ordinary decoding baselines (OD1, OD2) for reference. *Left*: cumulative unique clusters against run number for the prompt “bridal dress design ideas.” *Right*: density estimates of per-topic cluster counts over 50 topics and 50 runs. All six methods are clustered at $\tau = 0.73$ on each run’s first five ideas. The Deepseek-3 arms are re-run at temperature 1.0 (see text); the two simulated arms were not separately relevance-judged and are thinned with their real counterpart’s relevance ratio.

Appendix N Threshold Sensitivity: τ Sweep

Cluster counts depend on the cosine-similarity threshold τ , so we sweep $\tau \in [0.55, 0.95]$ on all three suites: the five public datasets ($\tau = 0.83$ in the main tables), the 50 high-stakes prompts and the 50 brainstorming topics (both $\tau = 0.73$). Figure A8 reports the

mean unique-cluster count for OD3 and RD3 using GPT-5.1. RD3 produces more clusters than OD3 at every tested threshold on every suite. Thus, the observed advantage of RD is robust to the choice of τ .

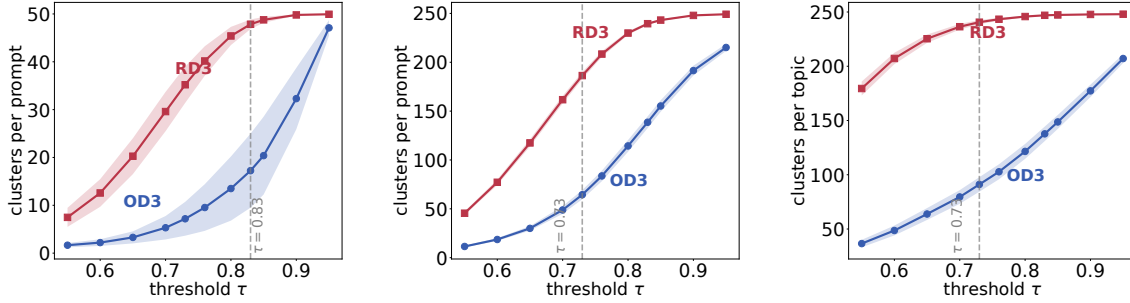


Figure A8: Threshold sensitivity. Mean unique-cluster count for OD3 (blue) and RD3 (red) as τ varies; bands are ± 1 standard error, dashed verticals the operating points used in the main paper. Left: five public datasets, 500 prompts, $\tau = 0.83$. Middle: 50 high-stakes prompts, $\tau = 0.73$. Right: 50 brainstorming topics, $\tau = 0.73$.

Appendix O Grammatical Correction Step

The grammatical correction is applied to both RD variants and all OD-based temperature-tuning methods, including the temperature-tuning ablation study in Appendix L. The prompt is below.

You are a strict grammar corrector, translator, and content filter. Follow these rules:

1. If the input is already in English and grammatically correct, return it ****unchanged****.
2. If there are grammar mistakes, correct them.
3. If the text is not in English, translate it into natural English.
4. If the content is ****nonsensical, gibberish, low-effort, or meaningless****, **CORRECT** them.

IMPORTANT:

- Do not explain or justify anything.
- Do not rephrase fluent English.
- Do not continue or expand.
- Output only the final corrected, translated, or filtered text — no commentary.

Here we provide four examples with respect to each model of raw outputs from RD before and after applying the grammatical correction step. The grammatical correction step improves the fluency of the outputs while preserving their diversity and creativity. The correction step does not change the semantic meanings of the outputs. Please note that GPT-5.1 fails to strictly follow the instruction. It tends to repeat the diverting tokens

in completing the next sentence. In comparison, Gemini-3.1-Pro has better instruction-following ability and directly generates fluent outputs.

Prompt: Respond in bullet points. Do NOT include sub-bullets. Limit each point to 10 words. Brainstorm 5 design ideas for a festival parade float.

Gemini-3.1-Pro (simulated completion through chat completions API):

Pre-correction:

- Decorate a giant wicker basket with colorful, oversized paper flowers. \n
- Honor past local heroes with large, painted portrait banners. \n
- Finish a massive surfboard sculpture with glossy, wave-like paint. \n
- Accessorize the float with shimmering streamers and vibrant confetti cannons. \n
- Spell out "LOVE" using giant, glowing, illuminated letters.

Post-correction:

- Decorate a giant wicker basket with oversized paper flowers. \n
- Feature local heroes with large, painted portrait banners. \n
- Finish a massive surfboard sculpture with glossy paint. \n
- Accessorize the float with shimmering streamers and confetti. \n
- Spell out "LOVE" using giant, illuminated letters.

GPT-5.1 (simulated completion through chat completions API):

Pre-correction:

- Non- Nonstop neon skyline with elevated train circling towering skyscrapers-
- Public garden oasis float featuring oversized blossoms and interactive pathways-
- Tem- Temporary youth hub float with art, music, and digital storytelling-
- Ups- Upside-down golf course float defying gravity with rotating greens-
- Qui- Quirky job fair float showcasing playful career-themed mini-exhibits'

Post-correction:

- Nonstop neon skyline with an elevated train circling towering skyscrapers -
- Public garden oasis float featuring oversized blossoms and interactive pathways -
- Temporary youth hub float with art, music, and digital storytelling -
- Upside-down golf course float defying gravity with rotating greens -
- Quirky job fair float showcasing playful career-themed mini-exhibits'

GPT-3.5 (real completion API):

Pre-correction:

- Elephants on stilts\n- Add stilts to elephants.\n- Combine concepts of walking and extravaganceperial elephants.\n- Watch for balance pand unstable situationscus\n- Gaze through the galaxy with extrobkea designs\n

Post-correction:

- Elephants on stilts for dynamic visual impact. \n
- Combine elements of walking and extravagance. \n
- Ensure balance to prevent unstable situations. \n
- Use outer space themes with galaxy designs. \n
- Incorporate swirling patterns for visual appeal. \n

Deepseek (real completion API):

Pre-correction:

Monsters and mythical creatures from around the world are coming to life and joining the parade in a celebration of global folklore and fantasy\n\nGiant dragon puppet winding through the streets\n- Floating unicorn island with sparkling waterfall mane\n- Phoenix rising from fiery nest on a float\n- Kraken tentacles emerging from a misty sea base\

Post-correction:

Monsters and mythical creatures from around the world are coming to life and joining the parade in a celebration of global folklore and fantasy:\n\nGiant dragon puppet winding through the streets\n- Floating unicorn island with sparkling waterfall mane\n- Phoenix rising from fiery nest on a float\n- Kraken tentacles emerging from a misty sea base\

Appendix P Computational Cost of RD vs. OD

We quantify the additional computation required by RD relative to OD. Each RD generation consists of two model calls: (i) an autoregressive decoding pass that receives the prompt, priming tokens, and diverting tokens and produces a raw response $Y_{RD,raw}$; and (ii) a grammar-correction pass that converts this response into the delivered output $Y_{RD,corr}$ (Appendix O). In contrast, OD requires only one decoding pass, producing Y_{OD} directly from the prompt P .

We measure cost per 1,000 delivered output tokens, where the delivered output is $Y_{RD,corr}$ for RD and Y_{OD} for OD. Define the RD-specific input overhead as

$$C \equiv |\text{priming}| + S \cdot |\text{diverting token}| + |\text{grammar}_{\text{sys}}|.$$

The input and output cost ratios of RD relative to OD are then

$$R_{in} = \frac{|P| + C + \bar{Y}_{RD,raw}}{|P|}, \quad R_{out} = \frac{\bar{Y}_{RD,raw} + \bar{Y}_{RD,corr}}{\bar{Y}_{RD,corr}} \approx 2.$$

The output ratio is approximately two because each delivered token is generated once during decoding and processed once by the grammar corrector. We use `cl100k_base` for tokenization.

Across six datasets and three models (GPT-3.5, GPT-5.1, and Gemini-3.1-Pro), RD incurs average ratios of

$$R_{in} = 13.6 \times \quad \text{and} \quad R_{out} = 2.15 \times .$$

The input overhead is dominated by the grammar corrector rereading the raw RD response, particularly when user prompts are short. Within the decoding pass itself, RD adds only

$$|\text{priming}| + S \cdot |\text{diverting token}| \approx 7\text{--}25$$

input tokens per generation. RD’s output overhead is also substantially lower than that of reasoning models, which commonly generate 10–20 $\times$ as many output tokens [OpenAI \(2024\)](#); [DeepSeek-AI \(2025\)](#).

The more relevant question for search is how many distinct ideas a method produces per unit of computation. To estimate this quantity, we combine the output-cost ratio $R_{\text{out}} = 2.15$ with the cluster counts reported in Tables [A5](#) and [A7](#), using the embedding-cosine metric over $T = 50$ runs. The resulting cluster yields per 1,000 delivered output tokens are:

Setting	RD3	OD3	RD3/OD3
Brainstorm ($\bar{Y}_{\text{corr}} \approx 150$)	14.9	12.2	1.22 $\times$
Public datasets ($\bar{Y}_{\text{corr}} \approx 300$)	1.48	1.15	1.29 $\times$

RD therefore produces more distinct ideas per delivered output token in both settings. Its 2.15 $\times$ output overhead is offset by an approximately 2.7 $\times$ increase in the number of distinct clusters, yielding a net advantage of about 1.22 $\times$ for brainstorming prompts and 1.3 $\times$ for public-dataset generations. More generally, the per-token yield ratio equals the cluster-count ratio divided by R_{out} . Thus, RD improves diversity per delivered token whenever its cluster count exceeds OD’s by more than 2.15 $\times$.

If cost is instead normalized by total input and output tokens, the corrector’s input pass increases RD’s effective overhead from 2.15 $\times$ to approximately 3.2 $\times$. Under this normalization, RD is roughly at parity with or slightly worse than OD in both settings. We use output-token normalization because it better reflects API pricing, where output tokens typically cost 3–5 $\times$ more than input tokens.

Appendix Q Human Evaluation of Relevance Judge

To validate the alignment between our LLM-based relevance judge and human judgments, we sampled 500 responses, stratified across methods and models, and collected human annotations via Amazon Mechanical Turk. The sample spans the same 50 brainstorming topics used in Section [4.5.3](#) at roughly ten responses per topic, and is balanced between decoding methods: 248 RD responses (across four RD variants) and 252 OD responses (across four models). Each response was independently evaluated on the same three-level scale the LLM judge uses (Relevant, Partially Relevant, Irrelevant). Surveys were fielded in 25 versions of 25 questions each; five questions per version were attention tests carrying a deliberately off-topic response, and a rater was retained only on a perfect attention score. Every submitted assignment cleared that bar, and six of the 25 surveys drew a fourth or fifth submission, so each response carries three to five independent ratings. Where a statistic requires a fixed number of raters, we use the variable-rater generalization ([Fleiss, 1971](#)). This human-subjects research was reviewed and approved by the Harvard University Committee on the Use of Human Subjects (the University Area Institutional Review Board) under protocol IRB25-0151, *Towards Diverse Text Generation in LLMs*. Informed consent was obtained from all participants after the nature and possible consequences of the study were explained, and no personally identifying information was collected.

Table [A10](#) reports the results. Among the 108 responses the LLM judge rated Partially Relevant, pairwise inter-rater agreement was only 73%, against 90% for the 381 responses

it rated Relevant: raters agree readily on what the judge already considers conventional, and disagree on what it flags as marginal. Collapsing those two categories into a single Relevant class raises majority-vote agreement with the judge from 80% on the three-level scale to 98%, with Fleiss’ $\kappa = 0.50$ (moderate agreement). Most of the residual three-level disagreement is therefore the judge’s Relevant/Partially Relevant boundary rather than a dispute about usability.

Two further patterns bear on the typicality bias documented in Appendix B. First, the judge applied the Partially Relevant label to 33.5% of RD responses but only 9.9% of OD responses—a factor of 3.4 on identical topics under an identical rubric. Second, human majorities do not sustain that distinction: they re-rate 77% of all Partially Relevant responses, and 73% of the RD ones, as fully Relevant. Were RD’s outputs genuinely more marginal, the raters would be expected to agree with the downgrade; instead they reverse most of it, which is what separates an instrument effect from a real difference in relevance.

Table A10: Human validation of the LLM relevance judge, on 500 responses drawn from the 50 brainstorming topics. Pairwise inter-rater agreement is pooled over rater pairs (agreeing pairs divided by total pairs) within each subset. The three-level majority-vote rate excludes the eight responses on which the raters split evenly ($n = 492$); collapsing to the binary scale leaves no ties, so that row uses all 500. The flip-up denominator is all 108 Partially Relevant responses, including the three of those eight that fall in this subset. Reproduced by `code/human_eval_agreement.py` in the replication archive, which writes the per-response labels and per-worker attention scores to `human_eval/analysis/`.

Quantity	Value
Responses evaluated	500 (248 RD, 252 OD)
Assignments collected / retained	81 / 81
Ratings per response	3–5
<i>Pairwise inter-rater agreement</i>	
LLM judge = Relevant ($n = 381$)	89.5%
LLM judge = Partially Relevant ($n = 108$)	73.1%
<i>Majority vote vs. LLM judge</i>	
Three-level scale	80.1% (394/492)
Binary (Relevant + Partially Relevant)	97.8% (489/500)
Fleiss’ κ , binary	0.50
<i>Partially Relevant label, by decoding method</i>	
Share of RD responses so labeled	33.5% (83/248)
Share of OD responses so labeled	9.9% (25/252)
Re-rated fully Relevant by human majority (all)	76.9% (83/108)
Re-rated fully Relevant by human majority (RD)	73.5% (61/83)

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